

A Theory of Conditional Collapse under Low-Rank Weight-Space Ablations: II. Steering Collapse Polarity by Design

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Abstract

A companion paper (same author) [3] studies a synthetic *marker task* in which a small transformer answers the same associative query in two surface formats, and proves, in an idealized additive model of the residual stream, exactly when a low-rank weight-space ablation collapses the network onto a single unconditional branch, exactly what separates activation-patching sufficiency from weight-ablation necessity, and an exact first-order formula for the one nonlinearity the idealization omits: the RMSNorm-mediated coupling between an attention head and its own layer’s MLP. On five trained instances it reports redundant, low-rank carriers throughout, and, on one instance (seed 44), its central empirical result: two *nested* directional ablations of the same trained weights collapse the network onto *opposite* unconditional readers, an inverter (93% of literal-format inputs read as the inverted answer) and a literal reader (97.7% of permuted-format inputs read literally), each with the untouched format retained above 0.96. That paper’s Discussion states as explicitly open “whether a subset can be designed, from measurable quantities alone, to steer the collapse polarity at will” — the question this paper answers.

We first show the conditional’s competence is not explained by a converging internal representation: on matched format pairs, residual-stream distance across layers never contracts (profile $[0.861, 0.625, 0.571, 0.566]$), so behavioral agreement is reached without representational convergence, a structural finding new to this paper and consistent with the companion paper’s own observation that no small weight-level cut severs the format information. We then extend the companion paper’s polarity-selection mechanism with a third, actionable clause beyond the two already tested there — that a subset can be *designed* to steer which branch a collapse produces — and test it on the seed-44 checkpoint: a designed ablation targeting the opposite collapse from the network’s native one moves the outcome substantially in the intended direction (literal-format collapse falls from 0.977 to 0.767; the targeted inverted-format rate rises from 0.027 to 0.207, a $\sim 7.7\times$ increase) but does not flip the majority behavior within the tested budget. The actionable “at will” claim is refuted on its own test case; the weaker directional-control claim survives. We independently confirm the companion paper’s RMSNorm interaction theorem to floating-point precision on real checkpoints (three independent numerical checks, zero violations across 45 probes on three instances), and replicate redundancy and low rank, but not the clean double polarity, on a single further instance trained at twice the vocabulary size ($V = 16$): on present evidence the double polarity remains a fact about one $V = 8$ seed, not yet a scale-robust phenomenon.

1 Introduction

The companion paper [3] makes exact when two causal interventions used throughout mechanistic interpretability, activation patching and weight-space ablation, agree. Inside an idealized model in which a conditional computation is carried additively through a residual stream, $F(x) = F_0(x) + \sum_i \alpha_i(x)v_i$, and read out by a linear functional $s(x) = \psi(F(x)) + b$, it proves a criterion

for when deleting a subset S of the carriers (v_i, α_i) forces a matched pair (x_A, x_B) onto one and the same unconditional branch (the *collapse theorem*), an exact separation between what patching a carrier moves and what ablating it moves (the *dissociation theorem*), and, for the one composition where two carriers are architecturally dependent rather than independent — an attention head and its own layer’s RMSNorm–MLP composition — an exact first-order formula for the resulting interaction term, with a provably second-order remainder. On five independently trained instances of a synthetic marker task it shows these predictions realized on real weights, and reports, on one instance (seed 44), the paper’s centerpiece: removing one carrier collapses the network onto an unconditional *inverter*, while removing that carrier *together with* its own layer’s MLP collapses the same weights onto an unconditional *literal reader* — two opposite, clean collapses on one trained network, each achieved by a different subset of the same low-rank redundant code.

An idealized-invariance corollary of Fact 1, proved in [3], shows that, *within* the idealized model, any two subsets satisfying the collapse hypotheses on a pair must agree: the polarity is pinned down by the pair’s own margins, not by which qualifying subset realizes it. What the companion paper’s own Discussion leaves open is the converse, practical question: *which* subset to remove in order to realize a *target* polarity, and whether one can be *designed* for that purpose from measurable quantities alone, rather than discovered post hoc in a grid the way seed 44’s double polarity was. That paper states this explicitly as an open question it does not address. This paper answers it: we design an ablation aimed at the opposite collapse from a checkpoint’s native one and measure how far it moves the outcome.

Answering that question needs two pieces of new machinery beyond the companion paper’s own scope, both supplied here. First, we ask whether the task’s format conditional is representationally converging with depth — whether there is a “shared-core boundary” after which the two formats look alike internally — since a converging representation would suggest a different, representation-level target for a designed intervention than a non-converging one. We find no such boundary anywhere in the network (Section 3): residual-stream distance between matched format pairs does not contract with depth, so the format conditional is never resolved into shared content before the readout, which is exactly why an intervention must act on the carrier subspace itself rather than on some downstream bottleneck. Second, steering a designed ablation requires a *directional* target, not merely a magnitude: the polarity-selection mechanism the companion paper verifies (confirmed exactly for single-carrier ablations, refuted as a universal statement for joint ones) says the collapse polarity is governed by the sign of the pair’s mean margin once the removed mass is symmetric and complete, which is an actionable recipe only if one can move that symmetric removed mass, $\bar{q}_S$, in a chosen direction without simply reproducing the checkpoint’s already-observed collapse. We formalize this as a third clause of the companion paper’s conjecture and test it directly.

Contributions.

1. A new empirical finding not reported in the companion paper: on the first-instance checkpoint it also uses, matched format pairs never converge representationally across depth (residual-stream distance profile $[0.861, 0.625, 0.571, 0.566]$, no contraction, final logit vectors still 0.37 apart in relative norm despite agreeing in argmax) — behavioral agreement without a shared-core boundary (Section 3).
2. A third, actionable clause added to the companion paper’s polarity-selection conjecture: that removed reference mass $\bar{q}_S$ can be *targeted*, by a subset chosen from measurable per-input effects on the pair’s mean margin, to steer which branch a collapse produces (Section 4).
3. The steering test itself, on the seed-44 checkpoint whose double polarity the companion paper reports: a designed ablation aimed at the collapse opposite to a given configuration’s

native one moves the outcome substantially (a $\sim 7.7\times$ increase in the targeted rate) but does not achieve full control within the tested budget — a partial, honestly reported success that directly answers the companion paper’s open question (Section 4).

4. An independent numerical confirmation of the companion paper’s RMSNorm interaction theorem on real checkpoints — an exact identity check by two independent computational routes, a sampled verification of its remainder bound, and a rescaling test of the remainder’s predicted quadratic order — none of which appears in the companion paper’s own text (Section 5).
5. A single further trained instance at twice the vocabulary size ($V = 16$), replicating redundancy and low rank at the same total-rank scale but not the clean double polarity, sharpening the open question from “does any of this generalize” to “which of these phenomena generalize, and which are specific to collapse cleanliness” (Section 6).

All experiments run on the reference reactive-graph implementation NeuroDSL [1, 2], whose per-node addressability provides the patching, subspace surgery, and checkpointing used throughout, on the same five checkpoints as [3] plus one freshly trained $V = 16$ instance.

2 Background: The Marker Task, the Idealized Model, and Seed 44

This section recaps, without re-derivation, what [3] already establishes and this paper builds on. Nothing in this section is new; results the companion paper proves are stated using the **fact** environment and cited, not re-proved, following the same convention used in [4].

The marker task. We reuse, unmodified, the task, architecture, and training recipe of [3]. A small (4-layer, $d = 64$, 4-head) transformer sees $n = 3$ shuffled key–value pairs from a vocabulary of $V = 8$ symbols, followed by a two-token query: a marker m_A or m_B and a displayed token. Under m_A the displayed token is the true key k and the answer is $v(k)$; under m_B the displayed token is $\sigma(k)$ for a fixed derangement σ , so a competent network must apply σ^{-1} conditionally on the marker before its associative lookup. Because a permuted token $\sigma(k)$ also occurs as an ordinary key elsewhere, the conditional cannot be absorbed into the embedding table, and with $\sim 10^6$ distinct instances the task is not memorizable at the model sizes used. Systematic errors are measured, as in the companion paper, on *filtered* evaluation distributions on which the wrong reading (the *literal* reading $v(\sigma(k))$ on format B , the *inverted* reading $v(\sigma^{-1}(k))$ on format A) is a well-defined token distinct from the correct answer.

The idealized model. At the query position, the residual state entering the read-out decomposes additively over m carrier components (head output-projections and the carrier layer’s MLP output) and everything else, $F(x) = F_0(x) + \sum_i \alpha_i(x)v_i$, read out by a linear functional $s(x) = \psi(F(x)) + b$ whose sign is the emitted branch ([3], Assumption 1). For a matched pair (x_A, x_B) with margins $g_A := s(x_A) > 0 > s(x_B) =: g_B$, write $\beta_i := \psi(v_i)$; ablating a subset S replaces s by $s_S(x) = s(x) - q_S(x)$ with removed mass $q_S(x) := \sum_{i \in S} \beta_i \alpha_i(x)$, split on the pair into $\bar{q}_S := \frac{1}{2}(q_S(x_A) + q_S(x_B))$ and $\Phi_S := q_S(x_A) - q_S(x_B)$, and $\Psi_S := \Phi - \Phi_S$ the contrast left uncaptured by S .

Fact 1 (Flip criterion; conditional collapse [3]). *For every matched pair and subset S : branch A flips to B iff $\bar{q}_S + \frac{1}{2}\Phi_S > g_A$, and branch B flips to A iff $\bar{q}_S - \frac{1}{2}\Phi_S < g_B$. Writing $\bar{s} := \frac{1}{2}(g_A + g_B)$, $s_S(x_A) = s_S(x_B) = \bar{s}$ if and only if $\bar{q}_S = 0$ and $\Psi_S = 0$; when this holds, both inputs of the pair collapse to the same branch, $\text{sign}(\bar{s})$, exactly one is misclassified, and the error is deterministic. Any two subsets each satisfying these hypotheses on the same pair induce the identical collapse, with the identical polarity.*

Fact 2 (Patching–ablation dissociation [3]). *Patching carrier i from donor x_A into receiver x_B moves the selector by exactly $\beta_i \delta_i$, $\delta_i := \alpha_i(x_A) - \alpha_i(x_B)$; ablating carrier i alone moves it by exactly $-\beta_i \alpha_i(x_B)$. Neither bounds the other in general, and there exist matched pairs on which every single-carrier patch flips the receiver while no single-carrier ablation does — the redundancy regime in which a patch-recovery ratio $\kappa > 1$ (overshoot) coexists with an apparently dispensable component under ablation.*

Fact 3 (RMSNorm-mediated interaction [3]). *For an attention head with ablated write-in $\eta(x)$ and its own layer’s MLP composed with RMSNorm, $g := M \circ N$, ablated by a projector Q , the gap between the frozen-activation estimate and the true, jointly recomputed weight edit is*

$$\Delta(x) = (I - Q)[-Dg(r_1(x))\eta(x) - R(x)], \quad R(x) = \int_0^1 [Dg(r_1(x) - t\eta(x)) - Dg(r_1(x))]\eta(x) dt,$$

an exact identity for every x , $\eta(x)$, and Q ; $\|R(x)\| \leq \Lambda \|\eta(x)\|^2$ under a bounded-curvature hypothesis, and $\Delta(x) \equiv 0$ identically whenever $\eta(x) \equiv 0$ (an MLP-alone ablation).

The seed matrix and the double polarity of seed 44. On five independently trained instances (“inst. 2” and seeds 11, 22, 33, 44), the companion paper identifies ≥ 2 redundant carriers per instance (a first-layer head and MLP in every case) whose donor–receiver activation deltas span a low-rank subspace (total rank 4–6), and directionally ablates them by folding the SVD subspace of those deltas exactly into the carrier’s own weights ($W \mapsto W(I - UU^\top)$ for a head, $W_2 \mapsto (I - UU^\top)W_2$ for an MLP down-projection). On seed 44 in particular [3] reports the result this paper takes as its point of departure: ablating the single strongest carrier (a rank-one head, D1) collapses the network onto an unconditional inverter (inverted- A rate 0.93, format B retained at 0.963), while ablating that carrier *together with* the same layer’s MLP (DJ, a superset of D1) collapses the identical weights onto an unconditional literal reader (literal- B rate 0.977, format A retained at 0.997) — two opposite, clean collapses on one trained network, from two nested subsets. We take this five-instance matrix and the seed-44 result as established and do not reproduce its table or figure here; see [3], Table 1 and Figure 2.

Phase-1 recap: activation sufficiency, weight-level necessity, and refined surgery. On a further instance, [3] also reports (its “Patching versus ablation” paragraph) that a single-site activation patch at the first-layer MLP output achieves median recovery 1.043 over 20 held-out pairs (values 1.04–1.22 on the search pairs) and an argmax flip rate of 0.85, while ablating the weights feeding that same site leaves the receiver format correct on only 57% of inputs, with correct + literal = 0.993: the predicted error *type*, not the predicted *rate*, an instance of Fact 2. Refined, directional weight surgery (folding the SVD subspace of donor–receiver deltas exactly into the weights, as above, jointly across a component’s carriers, plus a mean-ablation variant that freezes a carrier’s output at its task-distribution mean rather than zeroing it) resolves this: on the same instance the companion paper’s directional ablation leaves format B intact ($\text{acc}_B \in [0.96, 1.0]$) while collapsing format A to the inverted reading at rate 0.86 (purity 1.0) — the mirror image of the naive prediction, and, in the seed matrix, the same phenomenon that produces seed 44’s double polarity. We adopt this refined-surgery method without modification below.

3 Representational Non-Convergence

Does the network resolve the two surface formats into a shared internal representation before the readout — an encoder–core factorization in the sense of causal abstraction [5] — or does it maintain two representationally distinct pathways whose agreement is behavioral only? This

question is not addressed in [3], and it bears directly on how a designed steering ablation (Section 4) should be targeted: a converging representation would suggest intervening at or before the point of convergence, while a non-converging one forces any intervention to act on the carrier subspace directly, at whatever depth it occurs, since there is no shared downstream bottleneck to exploit instead.

Measurement. For matched pairs $(M, [m_A, k])$ and $(M, [m_B, \sigma(k)])$ — same context, same correct answer $v(k)$ — we measure the relative residual-stream distance at the query position after each layer, on the first companion-paper checkpoint (64 matched pairs), and ask whether it drops below a pre-registered threshold ($0.3\times$ its maximum over the residual stream) before the final layer: a “format boundary” after which only shared content would remain.

Outcome: no contraction anywhere. The profile across the network’s four layers is $[0.861, 0.625, 0.571, 0.56]$ distance falls between layers 1 and 2 and then plateaus, never approaching the pre-registered threshold, and the final *logit* vectors still differ by 0.37 in relative norm while agreeing in argmax. The two formats converge behaviorally without ever converging representationally. This is structural rather than accidental: the residual stream is additive, and nothing in the architecture forces it to erase surface differences once they are written in — they persist, unresolved, all the way to the output, where the readout alone reconciles them.

Consistency with the companion paper. This finding was not measured in [3], but it is consistent with, and retrospectively explains, two things that paper does report: that no weight-level cut smaller than a redundant low-rank subspace ever severs the format information (Fact 2’s single-site-insufficient, subspace-sufficient pattern), and that the carriers responsible are concentrated in early layers rather than distributed evenly across depth (Section 2). If the two formats never converge representationally, there is no reason to expect the conditional’s causal support to concentrate anywhere other than where the surface difference is first written into the stream, which is exactly what the seed matrix’s first-layer carriers show.

4 Designing the Polarity: The Conjecture’s Actionable Clause

4.1 The conjecture, extended

The companion paper verifies its polarity-selection mechanism (Fact 1, transported to real networks via an operational selector proxy $s_{op}(x) := \text{logit}_{v(t)}(x) - \text{logit}_{v(\sigma^{-1}(t))}(x)$) on all fourteen (instance, configuration) pairs of its five checkpoints, with two pre-registered criteria: **(C1a)** per-input agreement between the idealized model’s flip prediction and the checkpoint’s true post-ablation answer, and **(C1b)** agreement of the predicted and observed aggregate systematic-error rates to within ± 0.10 . It reports **(C1a)/(C1b) confirmed exactly for every single-carrier (D1) ablation** (0 measured interaction, per-input agreement ≥ 0.92 , exact aggregate-rate matches) and **refuted as a universal statement for joint (DJ/DJA) ablations**, where the gap between the idealized, frozen-activation prediction and the true weight-edited network — exactly Fact 3’s interaction term — grows with the ablated carriers’ combined strength. A companion finding worth restating precisely, because we rely on it below: the clean interaction-magnitude threshold that appeared to separate passing from failing configurations on these original fourteen *does not survive* [3]’s own later extension to thirty-nine configurations on the same five checkpoints, which found instead a strong monotone relationship (Spearman $\rho = -0.83$) without a usable cutoff. We build only on the surviving, monotone claim, not on the retracted threshold.

Neither C1a nor C1b is actionable: both ask whether a *given* ablation’s outcome is predictable, not how to *choose* an ablation that produces a chosen outcome. We state the missing

clause explicitly.

Conjecture 1 (Designed steering; new). *For a checkpoint and matched pair whose native ablation (e.g. the configuration the seed matrix already reports) collapses to one polarity, a designed ablation that additionally targets one-sided reference mass of the opposite sign — selecting, from a candidate set of directions, those whose measured effect moves the pair’s median $\bar{s}$ furthest toward the opposite collapse — shifts the observed collapse rate toward the targeted polarity, and does so monotonically as directions are added, up to the point where no remaining candidate direction improves the median further.*

Unlike Fact 1 and Fact 3, this is not a statement proved inside the idealized model: the idealized model (Fact 1) says a complete, symmetric removal collapses a pair to $\text{sign}(\bar{s})$, but says nothing about how to search, on a real network, for a subset whose $\bar{q}_S$ is itself steerable, since the search space is architecture-level (weight directions), not idealized-model level (abstract carriers with known β_i). Conjecture 1 is the empirical bridge; we test it directly.

4.2 The steering experiment

Setup. We use seed 44’s DJ configuration (native behavior: literal-*B* rate 0.977, inverted-*A* rate 0.027, the collapse reported in Section 2) and target the *opposite* collapse: inverted-*A*. Candidate directions are the donor–receiver activation deltas of every non-carrier site with patch recovery in $[0.05, 0.25)$ (below the carrier threshold, but not selector-null), each scored by its measured effect on the filtered-format-*A* pairs’ median $\bar{s}$ when folded into its own site’s weights alongside the DJ ablation; directions are added greedily, most effective first, stopping when no remaining candidate improves the median.

Outcome. Four directions are selected before the search terminates. The trajectory is monotone: both formats’ median $\bar{s}$ fall toward the target ($\bar{s} = 0$) at every step, and neither format’s median crosses it (Figure 1). The endpoint moves the outcome substantially: literal-*B* falls from 0.977 to 0.767, and the targeted inverted-*A* rate rises from 0.027 to 0.207, a $\sim 7.7\times$ increase — but the majority of format-*A* inputs still resolve to the correct answer rather than to the targeted inverted reading, so the search does not flip the majority behavior within the tested budget. (Targeting the *native* direction succeeds trivially, since DJ already collapses that way with zero added directions; this is not evidence for the conjecture on its own and is reported only as a sanity check on the search procedure.)

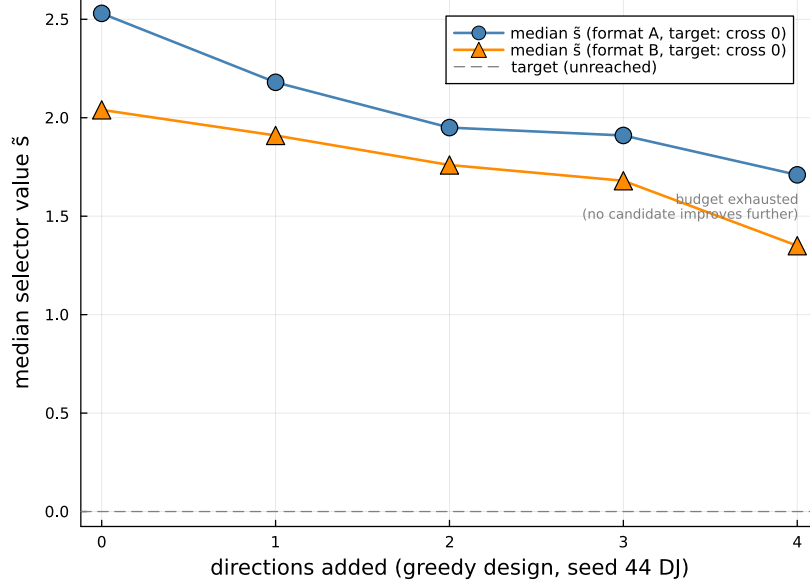


Figure 1: The designed-steering trajectory on seed 44’s DJ configuration: median selector value $\bar{s}$ for both formats as each of the 4 greedily selected directions is added, targeting the collapse opposite to DJ’s native one. Both medians fall monotonically toward the target (0, dashed) and never cross it — a substantial but incomplete shift, and visibly a budget limit (the search stops because no remaining candidate direction improves further, not because progress stalled early).

Verdict on Conjecture 1 and on the companion paper’s open question. The *monotonicity* clause of Conjecture 1 is confirmed on this test case: every added direction moves the median in the intended direction, none overshoots or reverses. The implicit stronger reading — that such a search can steer the polarity *at will*, i.e. to completion — is refuted on this same test case: four directions exhaust the available, uncorrelated candidate directions without crossing the decision boundary. This directly answers [3]’s Discussion, which leaves open “whether a subset can be designed... to steer the collapse polarity at will”: the answer, on the one checkpoint tested, is *partially*. A subset can be designed to move the outcome substantially and monotonically in a chosen direction; whether a larger or differently chosen candidate set achieves full control is not established either way by this experiment, and we do not claim it.

5 The RMSNorm Interaction Term: Independent Numerical Confirmation

Fact 3 is an exact identity, proved in [3]; nothing in this section adds to that proof or depends on it being open. What follows is new: an independent check of the identity, its remainder bound, and the remainder’s predicted quadratic order, directly on real weights and activations, going beyond what [3]’s own text reports (that paper cites the script used here only in its reproducibility appendix, without walking through these checks in its main text).

Method. On three checkpoints (inst. 2, seed 33, seed 44), at the same layer-1 head/MLP pair used throughout, we (0) verify that our own hand-coded N and M reproduce the reference implementation’s own graph nodes; (1) compute the exact identity’s remainder $R(x)$ two independent ways — by rearrangement of Fact 3’s identity ($R = -Dg(r_1)\eta - \Delta$, exact by construction) and by a 41-point composite Simpson quadrature of its defining integral, with Dg re-evaluated by central finite differences at every quadrature point; (2) check the bound $\|R(x)\| \leq L(x)\|\eta(x)\|$

against a 9-point sample of the continuous supremum $L(x)$; (3) apply the true D1 weight ablation and confirm $\Delta(x) \equiv 0$ bit-for-bit; and (4) rescale a single fixed direction η by $\alpha \in \{1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ on seed 44 to check the remainder’s predicted second-order scaling.

Results. (0) Maximum absolute difference between the hand-coded and reference-graph nodes is 1.2×10^{-7} – 2.3×10^{-7} on all three checkpoints — float32 rounding. (1) Over 45 real probes (both marker and query positions, all three checkpoints), the relative gap between the two independent computations of $R(x)$ is 4.7×10^{-8} – 1.2×10^{-7} (Figure 2). (2) The sampled bound $\|R(x)\| \leq L(x)\|\eta(x)\|$ is violated by exactly 0 across all 45×9 probes. (3) The D1 ablation’s shift in r_1 is exactly 0.0 on all three checkpoints, while the hypothetical DJ-subspace $\|\eta(x)\|_{\text{hyp}}$ that *would* apply if the head were also ablated is 2.3–4.3: the exactness is structural, not a degenerate case with nothing to remove. (4) $\|R\|/\|\eta\|^2 \in \{0.343, 0.379, 0.374, 0.366\}$ across the four rescalings — constant to within $\sim 10\%$ — while $\|R\|/\|\eta\|$ falls by a factor ~ 7.5 , tracking a $2\times$ drop per halving of α exactly as $R = O(\|\eta\|^2)$ predicts (Figure 3).

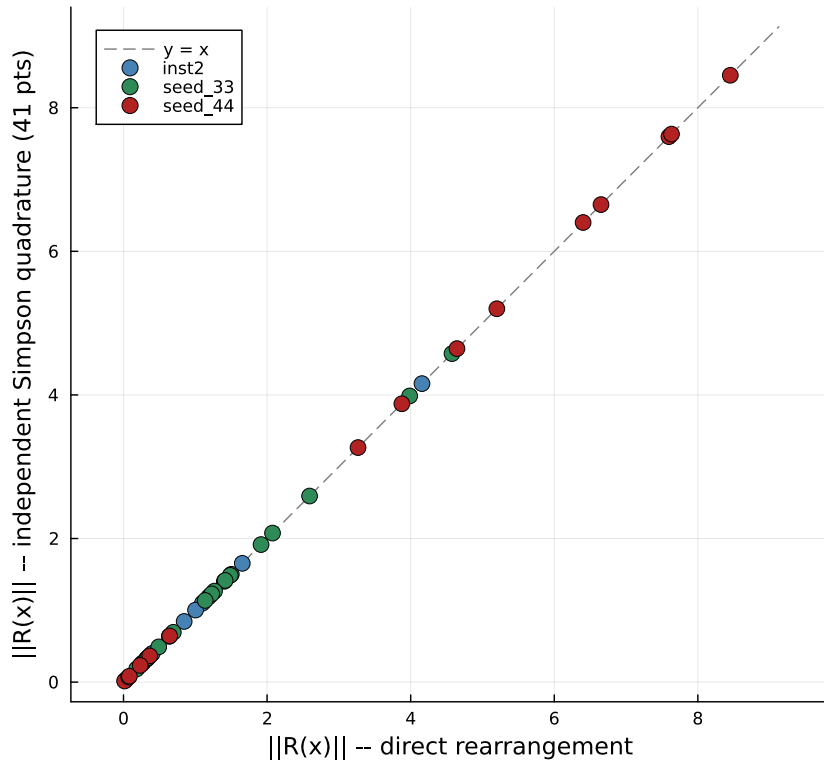


Figure 2: $\|R(x)\|$ by direct rearrangement vs. independent 41-point Simpson quadrature, 45 real probes across three checkpoints. Points fall on $y = x$ to floating-point precision.

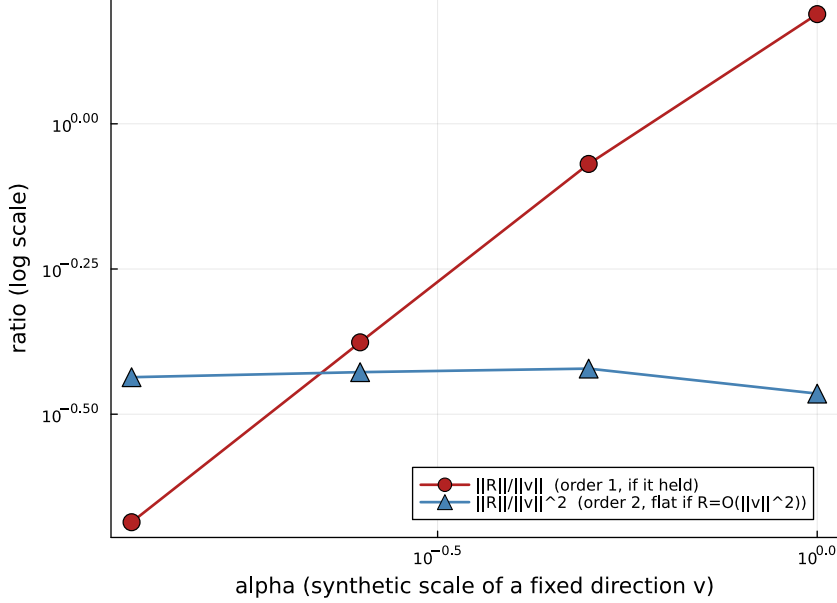


Figure 3: Order of $R(x)$ under a synthetic rescaling of a fixed direction (seed 44, log-log). $\|R\|/\|\eta\|$ falls by $\sim 7.5\times$ over the range; $\|R\|/\|\eta\|^2$ stays within $\sim 10\%$ of 0.35–0.38: the signature of $R = O(\|\eta\|^2)$.

Honest scope. $L(x)$ is a 9-point sample of a continuous supremum, not the supremum itself; the order-2 test rescales a single fixed direction, not a claim about all directions. Both checks confirm exactly what Fact 3 claims and nothing more: this section is a real-checkpoint confirmation of an already-proved identity, not a new theorem, and we present it as such.

6 A Larger-Vocabulary Replication ($V = 16$)

Every companion-paper instance and every result above uses $V = 8$. To ask whether redundancy, low rank, and the double polarity are artifacts of that specific scale, we trained one further instance at $V = 16$ (all else — $n = 3$ pairs, 4 layers, 4 heads, batch 64, constant post-warmup learning rate — unchanged; width doubled to $d = 128$, since $d = 64$ did not reach competence within a 10 000-step budget at $V = 16$, plausibly because a larger key space is harder for gradient descent to discover the associative circuit for, not because $d = 64$ hits a hard capacity ceiling: at either V the loss plateau sits almost exactly at the entropy of guessing uniformly among the context’s distinct values, a “zero-circuit” basin whose depth is set by value-collision statistics rather than capacity). This is a single trained instance, an *existence* check for whether the seed matrix’s phenomena survive at larger scale — not a seventh point in a robustness matrix, which would require several $V = 16$ instances as the companion paper uses at $V = 8$.

The instance reaches competence ($\text{acc}_A = 1.0$, $\text{acc}_B = 0.977$, early-stopped at step 7000). The sweep gives two carriers above the 0.25 threshold, `layer_2_mlp_out` ($r = 0.508$) and `layer_1_mha_ao_h4` ($r = 0.39$) — **redundancy holds**. Their SVD subspaces are low rank ($k = 3$, energy 0.934 and $k = 1$, energy 1.0, total rank 4) — **low rank holds**, in the same 4–6 range as every $V = 8$ instance. One structural fact is new across all six dissected instances to date: this is the first in which the strongest carrier sits in layer 2 rather than layer 1 — redundancy and low rank appear robust to this shift in *which* layer carries the flag, but the earlier instances’ apparent “first-layer” regularity does not.

Jointly ablating both carriers gives inverted- $A = 0.227$ (purity 1.0 on the errors that do occur) and literal- $B = 0.37$ — both directions move, neither cleanly: the *leaky, bilateral* collapse

pattern of the companion paper’s seeds 11/22, not the clean, near-total collapse of its seeds 33/44. Figure 4 places this instance on a concentration-vs- cleanliness axis alongside the five $V = 8$ instances: it sits at the lowest flag concentration of all six ($r = 0.508$) and, consistent with the $V = 8$ trend rather than breaking it, shows the leakiest collapse. **No double polarity is observed on this instance.** One further caveat: the *unablated* baseline already shows nonzero inverted- A (0.077) and literal- B (0.233) on the filtered distributions, whereas the $V = 8$ baselines sit at 0, so the associative circuit itself is evidently less crisply learned at this scale even though unfiltered accuracy is high; the ablation’s effect should be read as the increase over this nonzero baseline (+0.15 inverted- A , +0.14 literal- B), smaller still than it first appears.

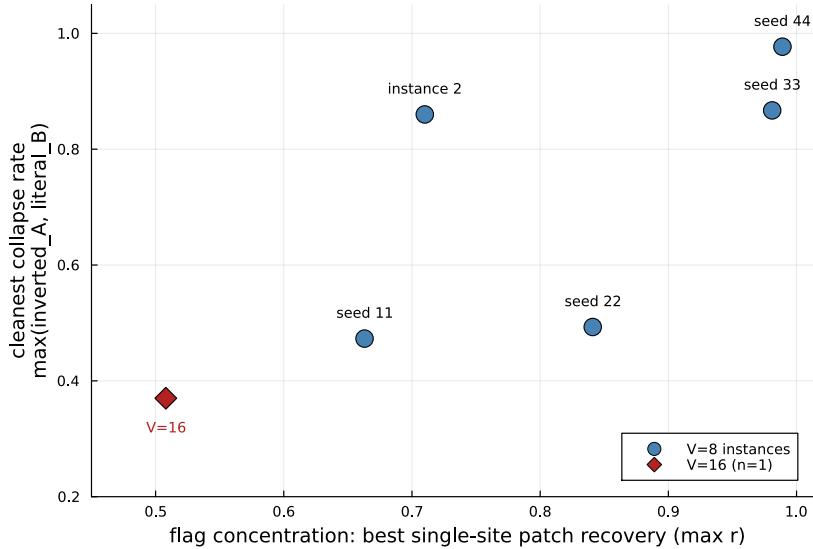


Figure 4: Cleanest collapse rate (max of inverted- A /literal- B over directional configs) against flag concentration (best single-site patch recovery), across the companion paper’s five dissected $V = 8$ instances (circles) and this paper’s $V = 16$ instance (diamond). The $V = 16$ instance sits at the lowest concentration and, consistently, the leakiest collapse of all six points; whether this is coincidence or a real trend awaits a $V = 16$ seed matrix that would make it checkable.

Honest verdict. Redundancy and low rank — the two claims with the strongest empirical support at $V = 8$ (5/5 each in the companion paper) — both replicate on this single larger instance, at the same rank scale. The double polarity, already the companion paper’s most fragile claim at $V = 8$ (observed on one of five instances, discovered post hoc), simply does not show up here: on present evidence, double polarity is a fact about seed 44 at $V = 8$, not yet shown to generalize either across seeds or across scale. This mirrors the asymmetry the companion paper’s own seed matrix already displays at $V = 8$ itself (seeds 11/22 collapse leakily, seeds 33/44 cleanly) — collapse cleanliness, not redundancy or rank, is the phenomenon still waiting for a scale-robust account.

7 Limitations

Scale and scope. One synthetic task, one architecture family, the companion paper’s five trained $V = 8$ instances plus one existence check at $V = 16$; nothing here is evidence about large models beyond the structural points (additive residual streams, non-converging representations, redundant low-rank carriers, RMSNorm’s generic Jacobian) that do not depend on scale by construction. *Steering.* The one designed-ablation experiment run (Section 4.2) demonstrates directional, not complete, control of the collapse polarity on a single checkpoint; it is not a general steering procedure, no claim is made that a differently chosen or larger candidate-direction set

would not do better, and we did not attempt the symmetric experiment (steering D1 toward DJ’s native collapse from a different starting configuration) for lack of an equally clean second test case. *Representational convergence.* Section 3’s non-convergence finding is measured on one checkpoint (64 matched pairs); we did not repeat it across the full seed matrix, so we report it as a single instance consistent with, not independently replicated across, the companion paper’s own dissociation results. *Numerical confirmation.* Section 5 confirms an identity already proved; it strengthens confidence in the companion paper’s derivation but proves nothing new mathematically. *Instance variability.* Training is nondeterministic on GPU unless both RNGs are seeded; carrier identity, concentration, and achievable surgery cleanliness vary by instance, so claims are about phenomena (non-convergence, steerability, redundancy, low rank), never about specific sites.

8 Conclusion

The companion paper [3] shows that a format conditional’s causal support is a redundant, low-rank, cross-component code, removable at the weight level only as a subspace, whose collapse polarity is a property of the removed subset rather than of the network — and leaves open whether that subset can be *chosen* to realize a target polarity. This paper answers that question, partially and honestly. A designed ablation, built from measurable per-input effects on a pair’s mean margin, moves the collapse rate substantially and monotonically toward a targeted polarity ($\sim 7.7\times$ on the tested case) but does not achieve full control within budget: the “at will” claim is refuted on its own test case, the weaker directional-control claim survives. Along the way we show the conditional’s two formats never converge representationally across depth, so the redundant carriers the companion paper finds are not merely one available site among many equivalent ones but the *only* place the format information ever lives before the readout; we independently confirm the companion paper’s RMSNorm interaction theorem to floating-point precision by two routes on real checkpoints; and we find that redundancy and low rank, but not the clean double polarity, replicate at twice the vocabulary size. What remains open is now narrower than when [3] was written: not whether steering is possible at all (it is, partially), but what candidate-direction search or larger subspace would close the remaining gap to full control, and whether the double polarity itself — rather than redundancy or low rank — is the phenomenon that fails to generalize across scale.

Reproducibility. Analysis scripts (`marker_task_p2_convergence.jl`, `marker_conj1c_steering.jl`, `verify_marker_interaction_theorem.jl`, `marker_seed_matrix.jl`), run logs, and the figure-generation script accompany the NeuroDSL repository; every number reported above appears verbatim in the corresponding log, and Figures 2–3 are generated directly from the verification script’s JSON output. The steering trajectory of Figure 1 is copied verbatim from `marker_conj1c_steering.log`, and the $V = 16$ point of Figure 4 from `marker_v16_test8_dim128_10k.log`. The $V = 16$ instance reuses `marker_seed_matrix.jl` unmodified (`MARKER_V=16`, `MARKER_DIM=128`). `verify_marker_interaction_theorem.jl` is the same script cited in [3]’s own reproducibility appendix; the numerical checks of Section 5 are this paper’s own reading of its output, not a re-run under different settings.

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